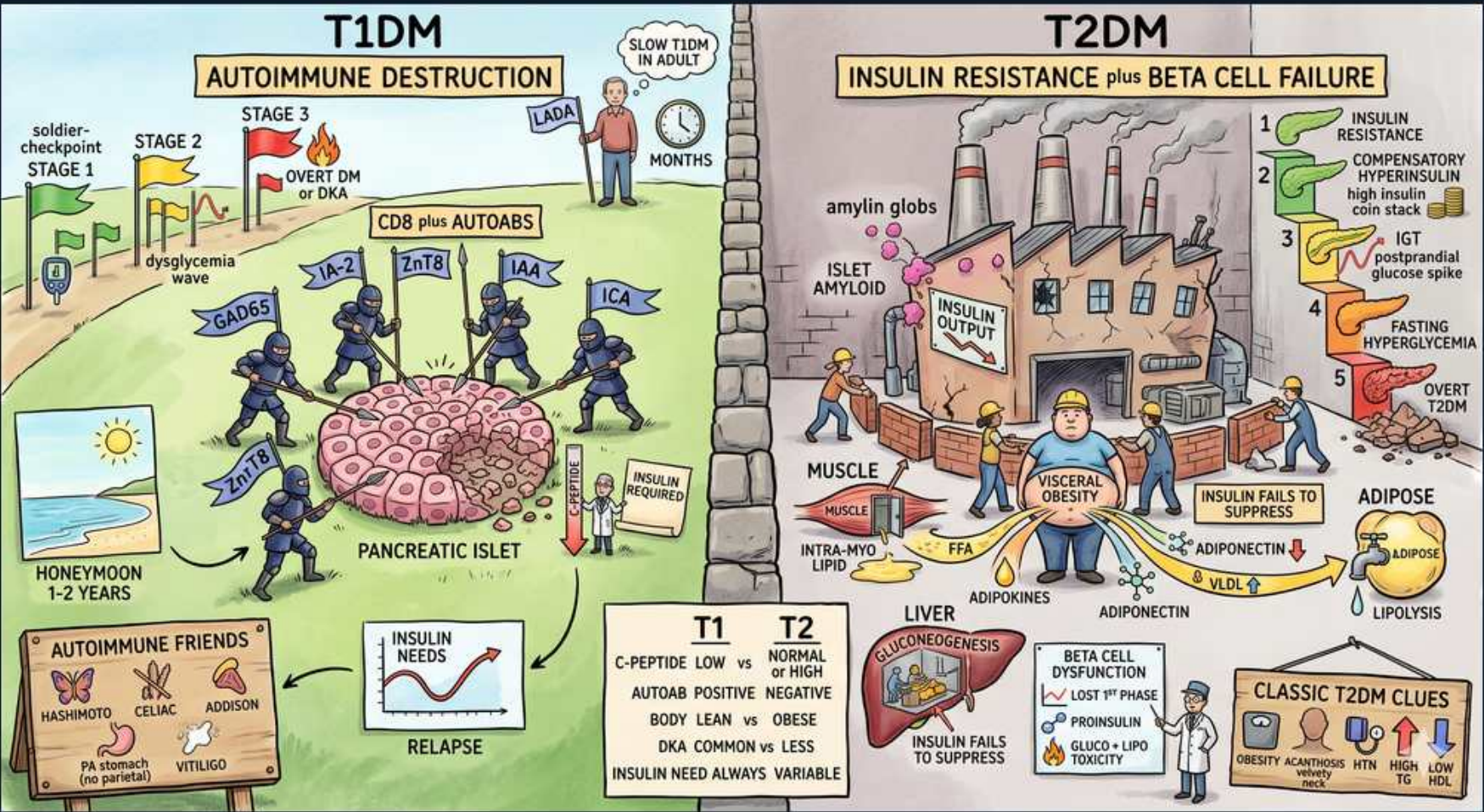


DM — Scene 2 — DM Pathophys & Dx: T1DM Battle vs T2DM Factory



1. Autoimmune destruction banner

WHAT YOU SEE

Banner T1DM AUTOIMMUNE DESTRUCTION

WHAT IT ENCODES

T1DM is a T-cell-mediated autoimmune disease: autoreactive CD8+ cytotoxic T cells (with CD4+ help) destroy pancreatic beta cells, producing ABSOLUTE insulin deficiency. Strongest genetic link is HLA-DR3/DR4-DQ2/DQ8 (class II MHC). Histology shows lymphocytic insulinitis. Result: lifelong dependence on exogenous insulin and proneness to DKA. Mnemonic: 'Type 1 = T cells + Total insulin loss.'

BOARD TRAP

Wrong answer: 'T1DM is caused by insulin resistance.' Discriminator — insulin resistance with relative deficiency is the T2DM mechanism; T1DM is autoimmune beta-cell DESTRUCTION with absolute deficiency. The presence of autoantibodies and low C-peptide settles it as type 1.

2. Stages 1-3 flags

WHAT YOU SEE

Flags STAGE 1 (autoabs, normoglycemia), STAGE 2 dysglycemia, STAGE 3 overt DM or DKA

WHAT IT ENCODES

T1DM evolves through defined stages: Stage 1 = ≥ 2 islet autoantibodies but NORMAL glucose (asymptomatic, presymptomatic); Stage 2 = ≥ 2 autoantibodies WITH dysglycemia (impaired fasting/glucose tolerance) still asymptomatic; Stage 3 = symptomatic overt hyperglycemia, often presenting as DKA. The anti-CD3 antibody teplizumab can DELAY progression from Stage 2 to Stage 3. Mnemonic: 'antibodies → dysglycemia → disease.'

BOARD TRAP

Wrong answer: 'a child must have symptomatic hyperglycemia to be considered to have type 1 diabetes.' Discriminator — Stages 1–2 are antibody-positive but normoglycemic/dysglycemic and asymptomatic; T1DM autoimmunity precedes clinical disease by years. Screening high-risk relatives can catch presymptomatic stages.

3. Autoantibodies (GAD65, IA-2, ZnT8, IAA, ICA)

WHAT YOU SEE

Soldiers labeled GAD65, IA-2, ZnT8, IAA, ICA

WHAT IT ENCODES

The five T1DM islet autoantibodies: GAD65 (glutamic acid decarboxylase — most useful/persistent in adults and LADA), IA-2 (insulinoma-associated/tyrosine phosphatase), ZnT8 (zinc transporter 8), IAA (insulin autoantibody — best in young children, but unreliable once exogenous insulin given), and ICA (islet cell). ≥ 2 positive confirms autoimmune diabetes. Mnemonic: 'GIANT-Z — GAD, IA-2, IAA, anti-islet (ICA), ZnT8.'

BOARD TRAP

Wrong answer: 'order an insulin autoantibody (IAA) to confirm T1DM in an adult already on insulin therapy.' Discriminator — IAA cannot distinguish autoimmune from injected-insulin antibodies once exogenous insulin is started; in an adult, GAD65 is the antibody of choice. Pick GAD65, not IAA, in treated adults.

4. LADA / slow T1DM clock

WHAT YOU SEE

LADA tag with slow T1DM in adult clock, months

WHAT IT ENCODES

LADA (latent autoimmune diabetes in adults) = slowly progressive autoimmune beta-cell loss presenting in adulthood. Patients are often lean, antibody-POSITIVE (GAD65), with initially preserved C-peptide that declines over months–years, progressing to insulin dependence faster than typical T2DM. Frequently MISdiagnosed as T2DM. Mnemonic: 'LADA = type 1 in slow motion, wearing a type-2 disguise.'

BOARD TRAP

Wrong answer: 'label a lean, antibody-positive 45-year-old who fails oral agents quickly as ordinary type 2 diabetes.' Discriminator — lean adult + GAD65 positive + rapid oral failure = LADA, not T2DM. Recognizing it changes management toward earlier insulin and avoids exhausting beta cells with secretagogues.

5. Pancreatic islet destroyed

WHAT YOU SEE

Damaged pink pancreatic islet, INSULIN REQUIRED

WHAT IT ENCODES

Autoimmune destruction wipes out beta cells (alpha/delta cells relatively spared), causing absolute insulin deficiency — so exogenous insulin is REQUIRED for life and is the only therapy that prevents DKA and death. C-peptide is low/undetectable, reflecting absent endogenous insulin. Mnemonic: 'no beta cells = no insulin = needle for life.'

BOARD TRAP

Wrong answer: 'manage newly diagnosed T1DM with metformin and lifestyle.' Discriminator — T1DM is insulin-deficient and metformin (which targets hepatic glucose/insulin resistance) cannot replace absent insulin; these patients need insulin from diagnosis. Oral agents alone are inappropriate.

6. Honeymoon 1-2 years

WHAT YOU SEE

Sun beach HONEYMOON 1-2 YEARS, then relapse graph

WHAT IT ENCODES

The T1DM 'honeymoon' is a transient partial remission soon after diagnosis: once glucotoxicity is relieved by initial insulin, residual beta cells temporarily recover, insulin requirements drop (sometimes near zero), and control is easy for ~1–2 years. Then beta-cell loss completes and requirements rise (relapse). Mnemonic: 'a short honeymoon before the marriage to insulin gets demanding.'

BOARD TRAP

Wrong answer: 'a T1DM patient whose insulin needs fell to near zero a few months after diagnosis has been cured / can stop insulin.' Discriminator — this is the honeymoon phase, NOT remission; stopping insulin risks rebound hyperglycemia/DKA when residual beta cells fail. Continue (low-dose) insulin through the honeymoon.

7. Autoimmune friends (Hashimoto, celiac, Addison, vitiligo, PA)

WHAT YOU SEE

Box AUTOIMMUNE FRIENDS: Hashimoto, celiac, Addison, vitiligo, pernicious anemia

WHAT IT ENCODES

T1DM clusters with other organ-specific autoimmune diseases (part of polyglandular autoimmune syndromes): Hashimoto/autoimmune thyroid (most common — screen TSH/TPO), celiac disease (screen tTG-IgA), Addison disease (primary adrenal insufficiency), vitiligo, and pernicious anemia (B12 deficiency). Screen at diagnosis and periodically. Mnemonic: 'autoimmune comes in packs — check thyroid, gut, adrenal.'

BOARD TRAP

Wrong answer: 'unexplained fatigue/hypoglycemia and hyperpigmentation in a T1DM patient is just poor glycemic control.' Discriminator — new hypoglycemia, fatigue, hyperpigmentation, and salt craving in T1DM should raise concern for coexisting Addison disease (autoimmune adrenal insufficiency); check morning cortisol/ACTH, don't just adjust insulin.

8. T2DM resistance + beta failure banner

WHAT YOU SEE

Banner T2DM INSULIN RESISTANCE plus BETA CELL FAILURE

WHAT IT ENCODES

T2DM has TWO core defects: peripheral/hepatic INSULIN RESISTANCE plus progressive BETA-CELL FAILURE, producing RELATIVE (not absolute) insulin deficiency. Early on, beta cells compensate with hyperinsulinemia, but they gradually fail. Strong polygenic/lifestyle (obesity) basis; NOT autoimmune. Mnemonic: 'Type 2 = resistance + tired (not destroyed) beta cells.'

BOARD TRAP

Wrong answer: 'T2DM is due to autoimmune destruction of beta cells.' Discriminator — autoimmune destruction (with autoantibodies, insulinitis, low C-peptide) is type 1; T2DM is insulin resistance with relative deficiency, autoantibody-NEGATIVE, normal/high C-peptide early. Antibody and C-peptide testing differentiate them.

9. Compensatory hyperinsulinemia stack

WHAT YOU SEE

Stacked coins: insulin resistance -> compensatory hyperinsulinism -> IGT spike -> fasting hyperglycemia -> overt T2DM

WHAT IT ENCODES

Natural history of T2DM: insulin resistance → compensatory hyperinsulinemia (normoglycemia maintained) → beta cells can't keep up → postprandial spikes (impaired glucose tolerance, IGT) → fasting hyperglycemia (impaired fasting glucose) → overt T2DM. Postprandial/IGT changes precede fasting hyperglycemia. Mnemonic: 'resistance first, fasting glucose last.'

BOARD TRAP

Wrong answer: 'a fasting glucose of 100 rules out any glucose abnormality.' Discriminator — early T2DM shows postprandial/2-hr OGTT elevations (IGT) BEFORE fasting glucose rises; a normal FPG can coexist with IGT. Use an OGTT or A1c if you suspect early dysglycemia despite normal FPG.

10. Islet amyloid

WHAT YOU SEE

Amylin globs ISLET AMYLOID

WHAT IT ENCODES

The characteristic islet pathology of T2DM is amyloid deposition derived from islet amyloid polypeptide (amylin/IAPP), co-secreted with insulin; accumulation is toxic to beta cells and contributes to their failure. Contrast with T1DM's lymphocytic insulinitis. Mnemonic: 'Type 2 islet = amyloid; Type 1 islet = lymphocytes (insulinitis).'

BOARD TRAP

Wrong answer: 'lymphocytic insulinitis is the islet finding in type 2 diabetes.' Discriminator — insulinitis (lymphocytic infiltration) is the T1DM lesion; T2DM islets show amylin-derived AMYLOID. Mixing these up is a classic histology distractor.

11. Visceral obesity hub

WHAT YOU SEE

Central VISCERAL OBESITY figure feeding muscle/liver/adipose

WHAT IT ENCODES

Visceral (central) obesity is the engine of T2DM insulin resistance: excess visceral fat releases free fatty acids and pro-inflammatory adipokines (low adiponectin, high resistin/TNF/IL-6) that impair insulin signaling in muscle (reduced glucose uptake), liver (unsuppressed gluconeogenesis), and adipose (unrestrained lipolysis). Mnemonic: 'belly fat broadcasts FFAs and bad adipokines to every tissue.'

BOARD TRAP

Wrong answer: 'BMI/total body fat matters more than fat distribution for insulin resistance.' Discriminator — VISCERAL/central adiposity (waist circumference) drives insulin resistance more than subcutaneous fat or total BMI; a 'metabolically obese' normal-weight patient can be insulin resistant. Assess waist, not BMI alone.

12. Liver gluconeogenesis

WHAT YOU SEE

Liver GLUCONEOGENESIS, insulin fails to suppress

WHAT IT ENCODES

Hepatic insulin resistance: normally insulin suppresses hepatic gluconeogenesis/glycogenolysis, but in T2DM the liver keeps producing glucose despite hyperinsulinemia, driving FASTING hyperglycemia. This is precisely the step metformin targets (it reduces hepatic glucose output). Mnemonic: 'the resistant liver won't stop making sugar — metformin tells it to stop.'

BOARD TRAP

Wrong answer: 'fasting hyperglycemia in T2DM is mainly from dietary carbohydrate intake overnight.' Discriminator — fasting (especially morning) hyperglycemia reflects unsuppressed HEPATIC gluconeogenesis, not overnight food; that's why metformin (a suppressor of hepatic glucose output) preferentially lowers fasting glucose.

13. T1 vs T2 comparison box

WHAT YOU SEE

Box: C-peptide low vs normal/high, autoab pos vs neg, lean vs obese, DKA common vs less

WHAT IT ENCODES

Head-to-head: T1DM = LOW C-peptide, autoantibody POSITIVE, LEAN, DKA-prone, insulin always required, often younger/abrupt onset. T2DM = NORMAL/HIGH C-peptide (early), autoantibody NEGATIVE, OBESE, less DKA (more HHS), gradual onset, oral agents often suffice initially. C-peptide measures endogenous insulin (low = beta-cell failure of type 1). Mnemonic: 'C-peptide LOW = type 1; C-peptide HIGH = type 2.'

BOARD TRAP

Wrong answer: 'a high C-peptide indicates type 1 diabetes.' Discriminator — C-peptide is LOW in T1DM (no endogenous insulin) and NORMAL/HIGH in T2DM (insulin present but resistant). Reversing the direction is a frequent trap; pair C-peptide with autoantibodies to classify.

14. Classic T2DM clues

WHAT YOU SEE

Box CLASSIC T2DM CLUES: obesity, acanthosis, HTN, HLD, family hx

WHAT IT ENCODES

Clinical clues to T2DM: central OBESITY, ACANTHOSIS NIGRICANS (velvety hyperpigmented neck/axilla folds — a marker of hyperinsulinemia/insulin resistance), and the metabolic syndrome cluster — hypertension, atherogenic dyslipidemia (high triglycerides, low HDL), plus a strong FAMILY HISTORY (T2DM is more heritable than T1DM). Mnemonic: 'fat folds (acanthosis), high BP, high TGs, family — that's type 2.'

BOARD TRAP

Wrong answer: 'acanthosis nigricans in a child points to type 1 diabetes.' Discriminator — acanthosis nigricans reflects hyperINSULINemia/insulin resistance, a TYPE 2 (and prediabetes/obesity) finding, not type 1. Its presence argues against autoimmune insulin-deficient diabetes.